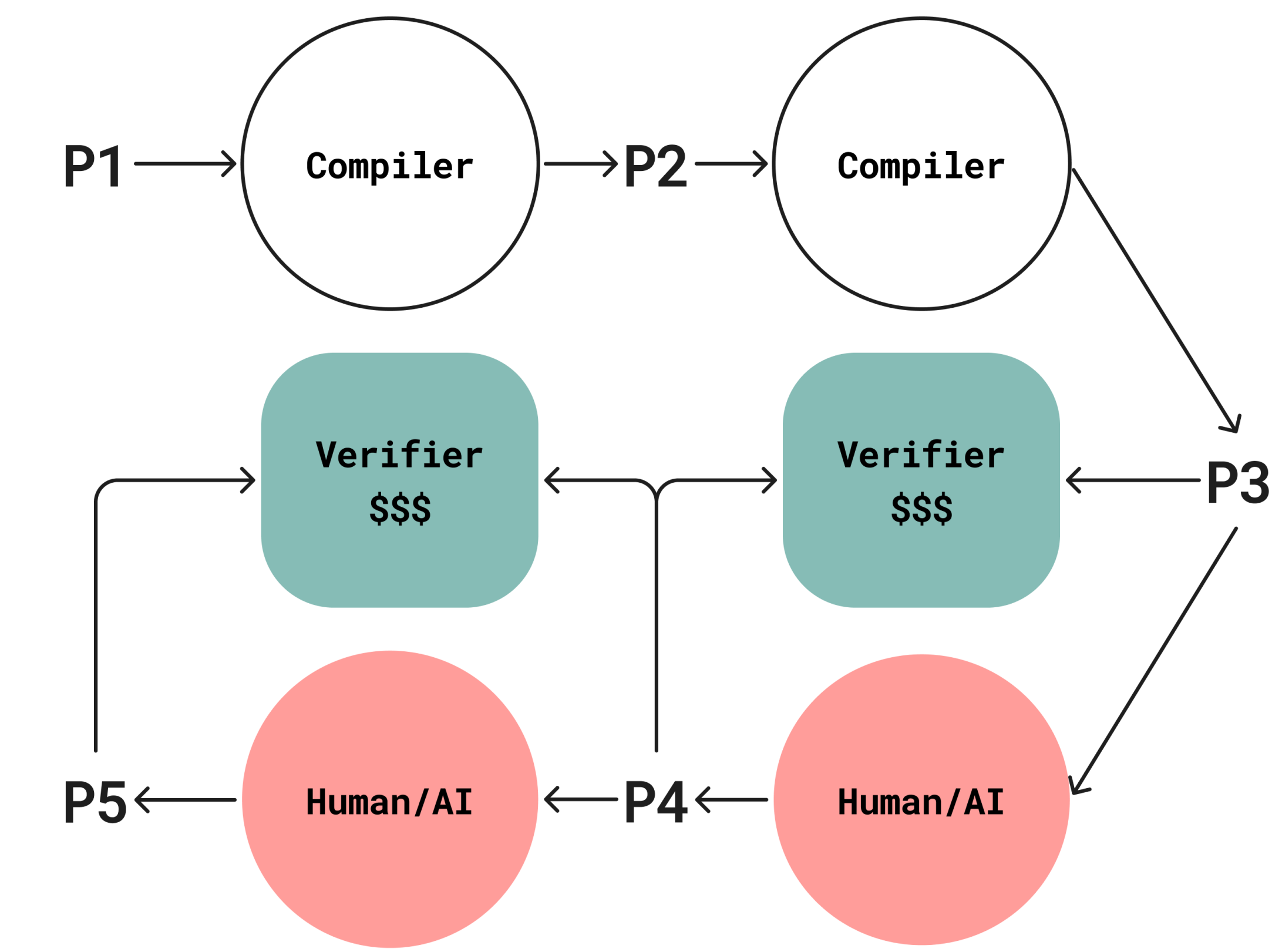
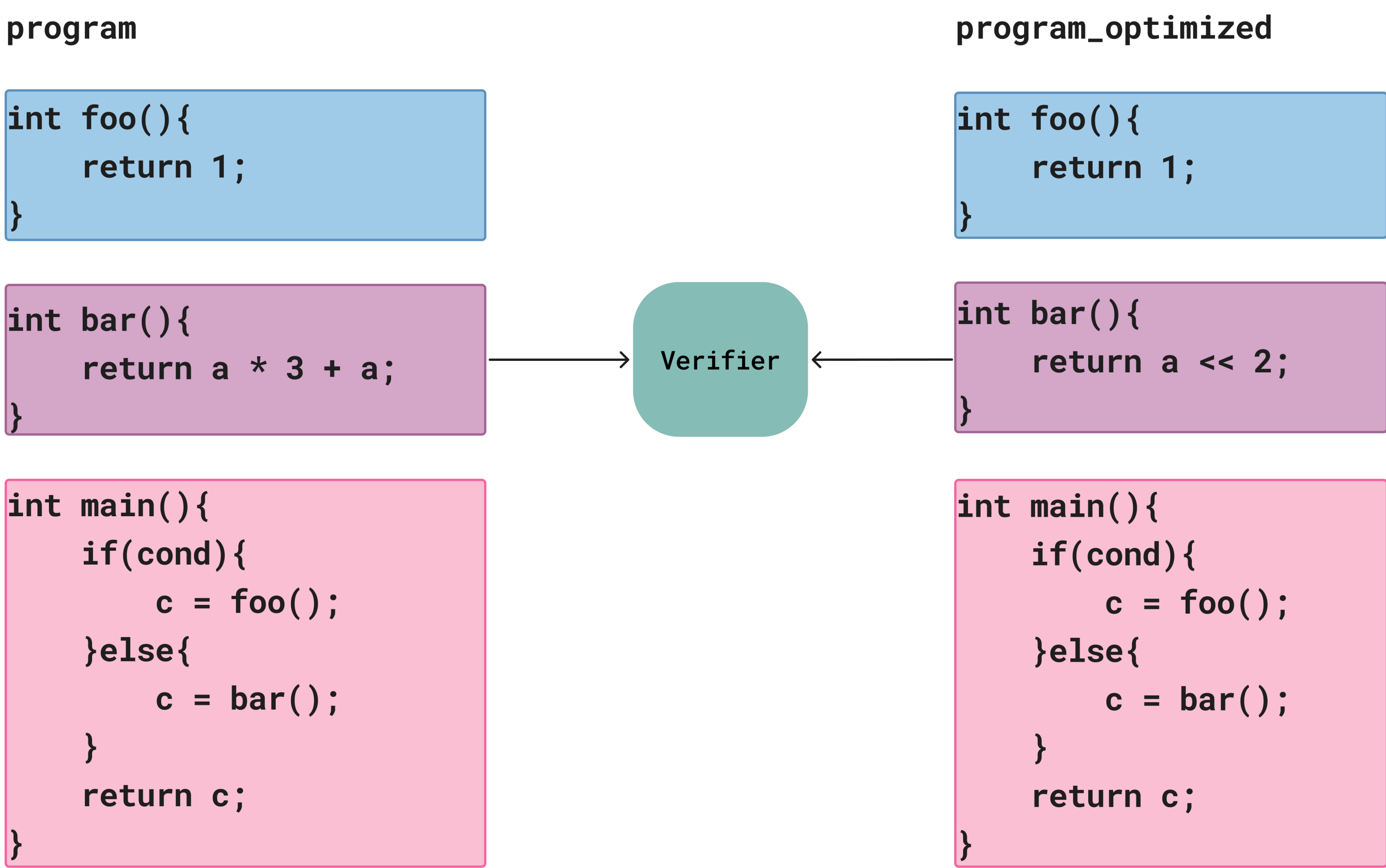


Motivation



An optimization pass by Human/AI is not as reliable as one by a time-tested compiler!

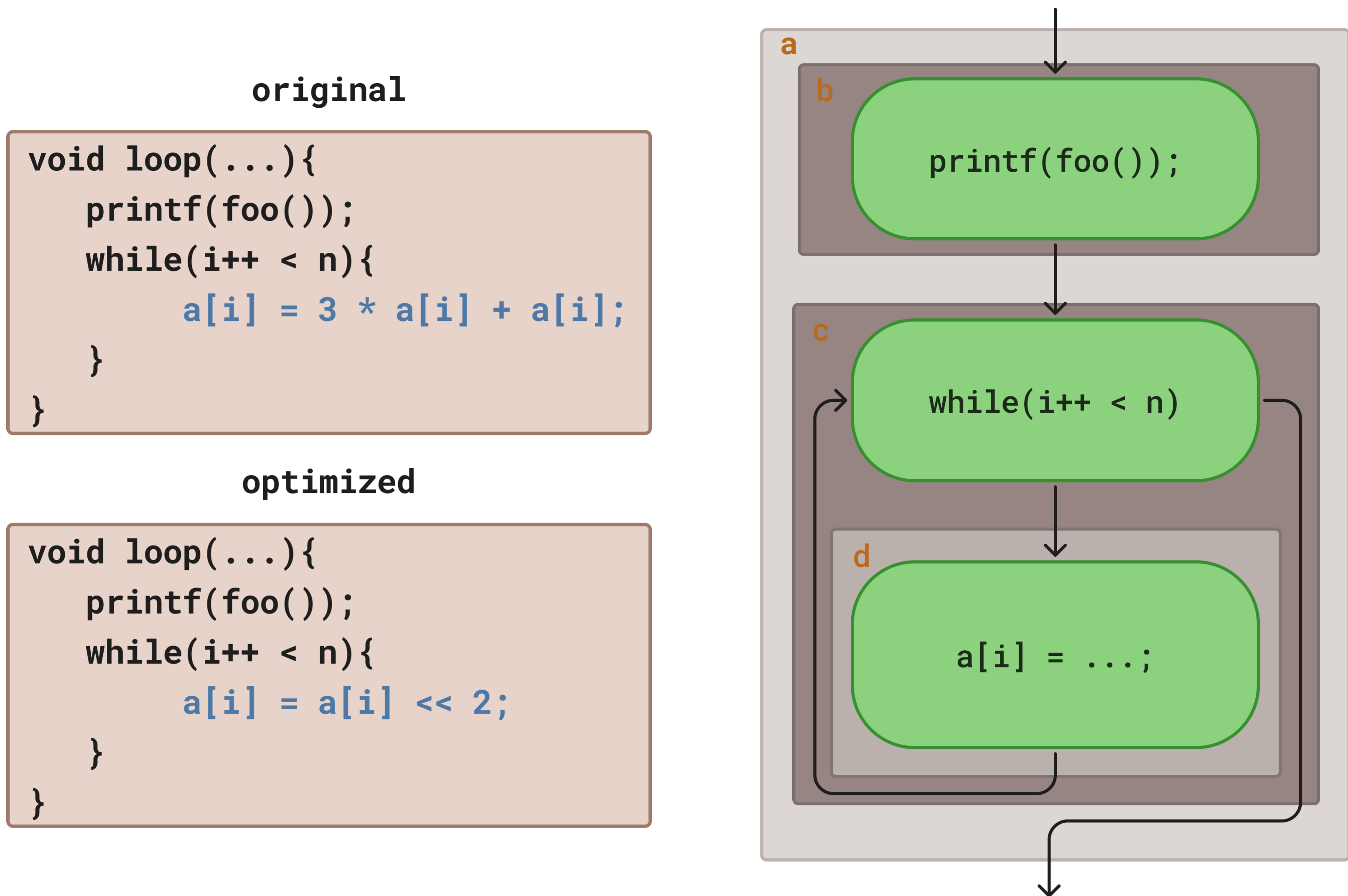
Background



SOTA: verification at the level of functions (compositional)

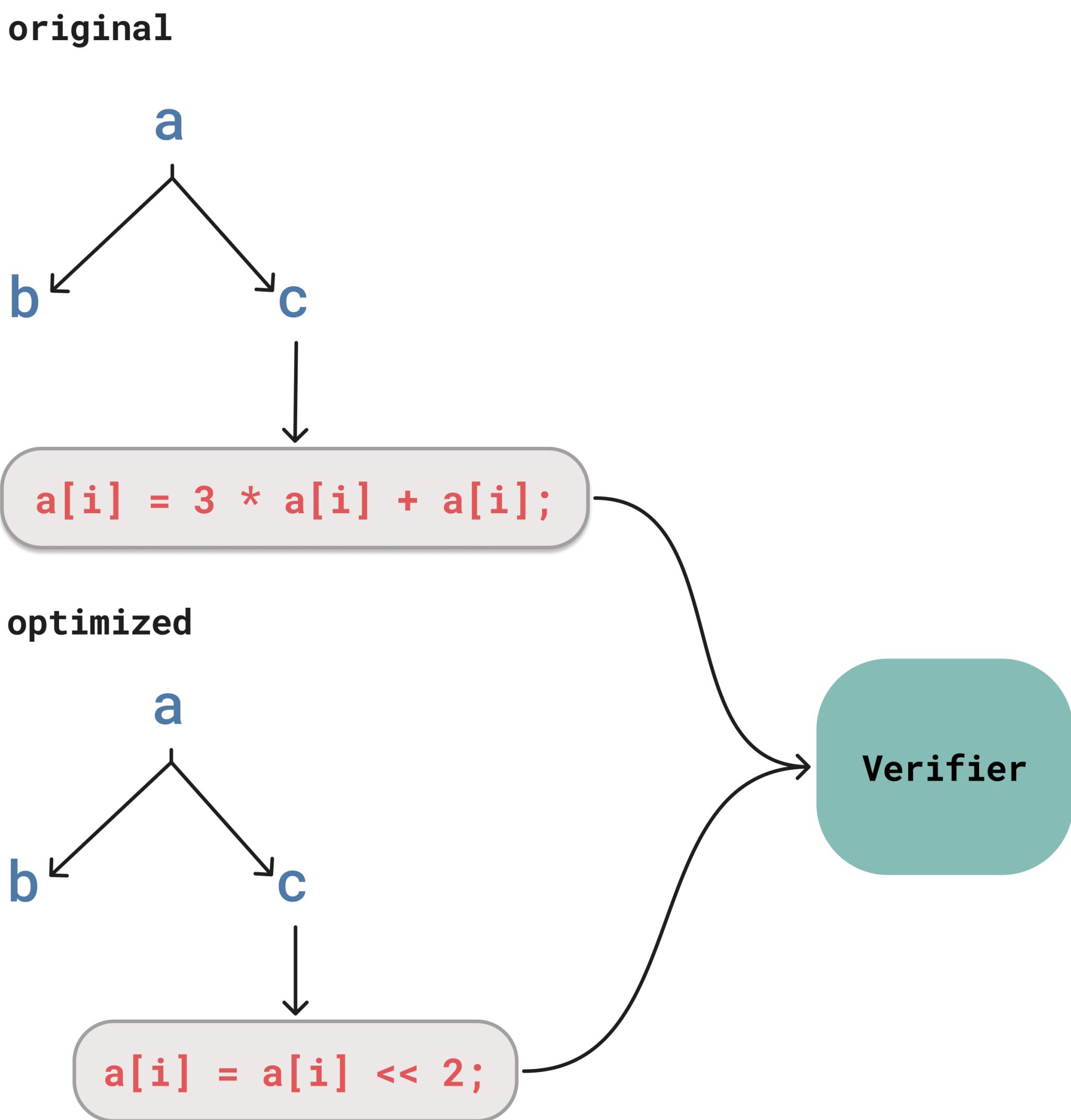
Finer granularity

Single Entry, Single Exit



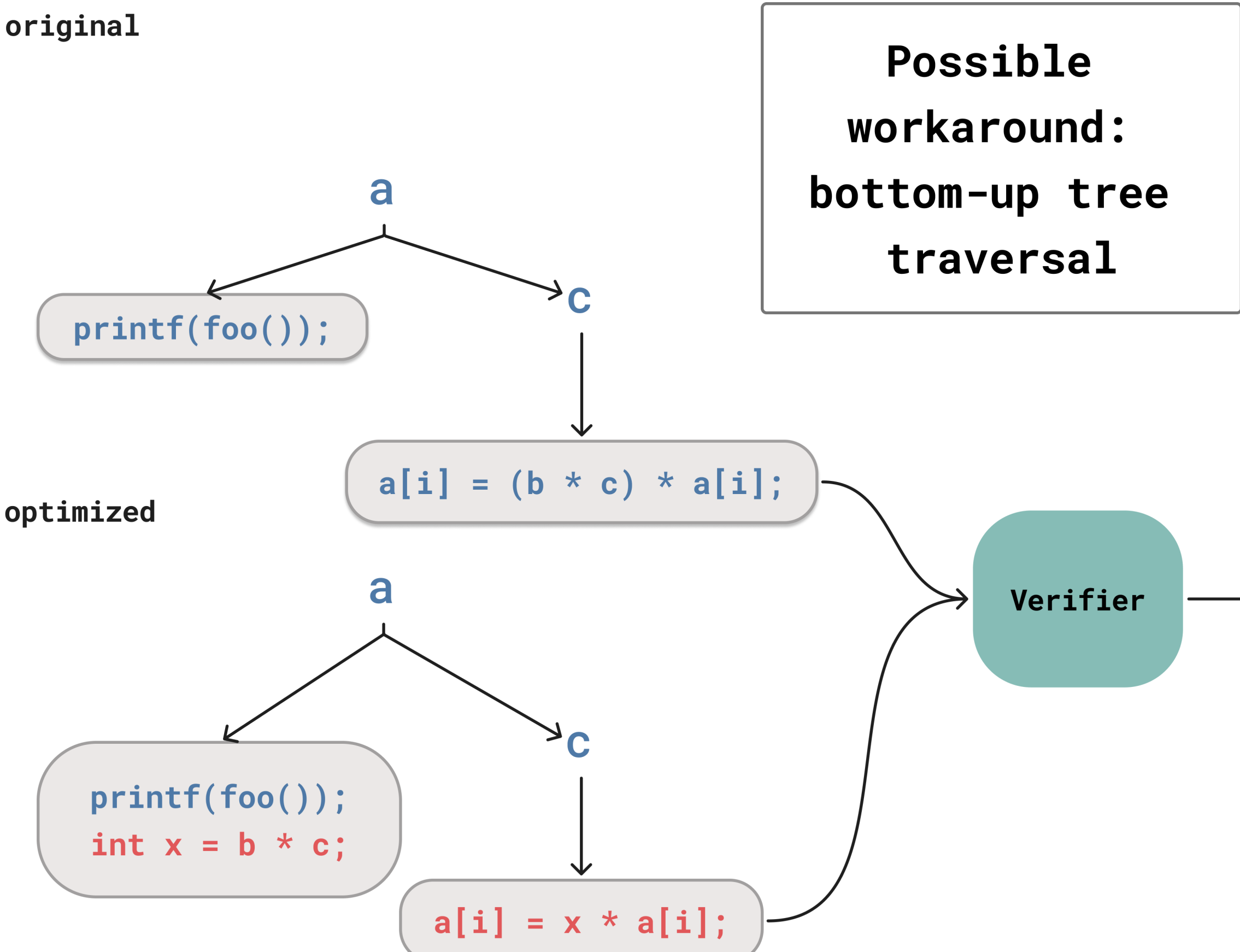
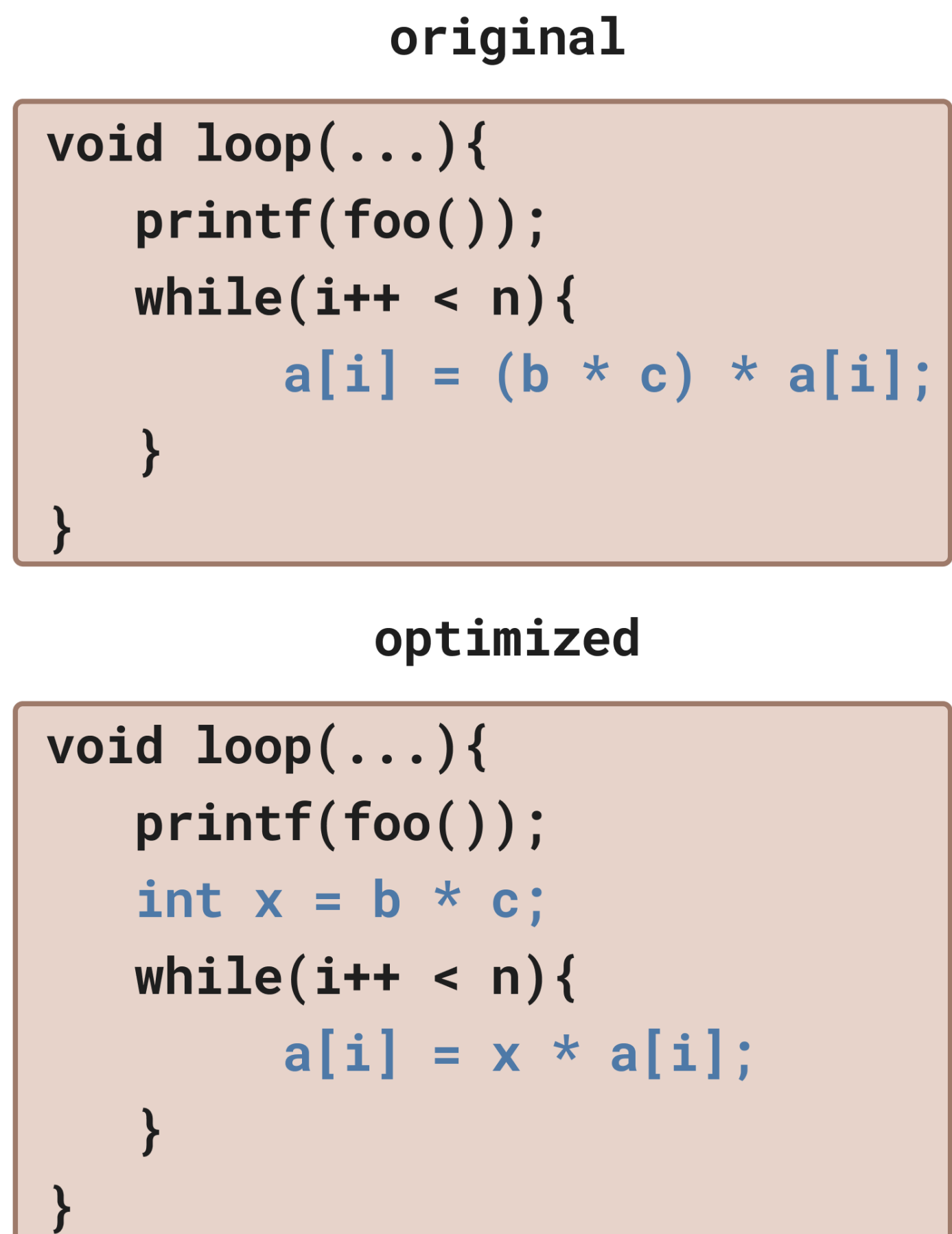
Optimizations within blocks are OK

Program structure graph



Verification at the level of blocks

Potential tradeoff: optimizations between blocks



Possible workaround: bottom-up tree traversal

Feedback

Do you see any additional tradeoffs with this approach?

Do you know of any other ways to break down the problem into smaller pieces?